

H2 Physics Prelim 2022 Solutions

- 1 (a) Thrust = resistive force since velocity is constant. C1
- Power = $Fv = Dv$
- $D \times 15 = 90000$
- $D = 90000/15 = 6.0 \text{ kN}$ A1
- (b) $F = \text{Thrust}$ Power = Fv
- $F - D = ma$
- $P/v - D = ma$
- $120000/15 - 6000 = 800a$ M1
- $a = 2.5 \text{ m s}^{-2}$ A1
- (c) As the velocity increases with time, the resistive force also increases . B1
- The driving force decreases as the speed of the boat increases since driving B1
- power stay constant, the net force on the boat decreases and hence acceleration B1
- decreases with time.
- Acceleration drops to zero when the resistive force is equal to the driving force on B1
- the boat.

- 2 (a) Gravitational force provides centripetal force. Same gravitational force acting on star